The two-cell reduction at the committed deck: the five-term concentration, the mixed-cell kill at scale $2m$, and the negative resolution of the strip-shape hypothesis (Round 28)

GST Hodge campaign — the operator's standing completion program

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Abstract

This round constructs the reduction demanded by every failed close of the campaign: AUDIT-22's finding F1 asked for the equivariant-descent spectral sequence carrying the traceless two-cell class $[C_T^0] \in H^2(\mathcal{G}_\bullet; \mathrm{SL}(H_z))$ to the deck computation; the 21-c item (ii) named the nonabelian-to-abelian reduction as the remaining construction. The construction delivered here, at the committed deck structure of the eight-letter world: **(T1)** the Lyndon–Hochschild–Serre five-term exact sequence for $1 \to \Gamma \to \mathcal{G}_\bullet \to C_4 \to 1$ with the traceless content module $A = \iota(M)$, with $E^{2,0} = H^2(C_4; A) = 0$ by the committed R1 law and $E^{0,2}$ computed exactly (rank 12, with the θ-shaped ι-strip part exactly zero), giving the *injectivity* of the mixed-cell map $H^2(\mathcal{G}_\bullet; A) \hookrightarrow H^1(C_4; \mathrm{Hom}(\Gamma, A))$; **(T2)** the mixed layer is computed exactly: $H^1(C_4; \mathrm{Hom}(\Gamma, A)) \cong (\mathbb{Z}/2)^6$ — pure 2-torsion, no $(x{+}1)(x^2{+}1)$-glue — so $2[C_T^0] = 0$: the doubled transport is cohomologically flat, and the rational halving of the two-center cycle closes; **(T3)** the strip-shape hypothesis (item (v)) is *inconsistent* — no nonzero ι-strip class is C_4-invariant, so the hypothesis is resolved negatively and replaced by the mixed-cell shape; **(T4)** the $(A, -A)$ witness: its mixed-cell class is nonzero at scale 1 (the obstruction lives) and dies at scale 2 (the parity engine again), and is *realized* by an explicit lifting two-cell cocycle c_ν (the d_2-leg discharged constructively - a lifting, not a retyping); **(T5)** the input-sensitivity is the Chern-typing barrier $ch_p(E) \in H^{p,p}$ (the Corollary-7 discipline localized), with the excluded-input run at the module level (the swap-typed content refused: its deck class survives); **(T6)** the item (iv) rank parity is *not needed* on this route — no eigenbranch exchange is required. The remaining boundaries are stated on the face: the torsion boundary $\partial \in H^3(\mathcal{G}_\bullet; \mu_2)$ for the nonabelian reconstruction, the fiber-centralizer condition, and the general-input deck structure (item (i) transformed). Certificate: `two_cell_reduction_checks.py`, 125 checks, exit 0, deterministic double-run byte-identical. Independent adversarial audit (AUDIT-28, 31 re-derivation probes, 0 disagreements): verdict PASS_WITH_REPAIRS_REQUIRED on the first pass (five receipt-level majors: the injectivity gated by a constant, the mixed structure printed not asserted, a tautological input receipt, the unpinned witness letter, the uncomputed d_2-leg); the batched repair round discharged all five with new gating checks (T1d, T2-assertions, T4-receipts, T4-LIFT, T5-excluded); the closing re-check confirmed.

1 Setting and the exact sequence

The committed deck structure. $\mathcal{G}_\bullet = \Gamma \rtimes C_4$ is the affine translation groupoid of the eight-letter world: $\Gamma = \mathbb{Z}^8$ the period lattice with letters e_j, f_j ($j = 0..3$) and the polarization $\theta = \sum_j e_j \wedge f_j$; the deck C_4 is the single-factor Gaussian rotation $g = (i, 1, 1, 1)$ (the committed convention: $e_0 \mapsto -f_0$, $f_0 \mapsto e_0$, the other six letters fixed), an exact element of $\mathrm{Sp}(8, \mathbb{Z})$ ($\gamma^T \theta \gamma = \theta$, certificate check

0003). The content module $A = \iota(M)$ is the traceless carrier content: the ι-image $\iota(x) = (x, -x)$ of the Weil plane $M = \mathbb{Z}\langle W_R, W_I\rangle$ with the deck acting by $\gamma(W_R) = -W_I$, $\gamma(W_I) = W_R$ ($\gamma^2 = -1$, 2-torsion-free; the ι-intertwining $(\gamma \times \gamma) \circ \iota = \iota \circ \gamma$ is the committed R3 receipt). At the cochain model of this round the fiber periods act trivially on the content ($A^\Gamma = A$); the honest boundary this idealization carries is recorded in §5.

The five-term sequence. For $1 \to \Gamma \to \mathcal{G}_\bullet \to C_4 \to 1$ with the $\mathcal{G}_\bullet$-module A, the Lyndon–Hochschild–Serre low-degree exact sequence reads [STANDARD STRENGTH, written out]:

$$0 \to H^1(C_4; A^\Gamma) \to H^1(\mathcal{G}_\bullet; A) \to H^1(\Gamma; A)^{C_4} \to H^2(C_4; A^\Gamma) \to H^2(\mathcal{G}_\bullet; A) \to H^1(C_4; H^1(\Gamma; A)) \xrightarrow{d_2} H^3(C_4; A^\Gamma).$$

This is the equivariant-descent spectral sequence demanded by AUDIT-22 F1, in its exact normalization at the degrees the two-cell lives.

2 The concentration (T1) [PROVEN HERE + CERTIFICATE RECEIPT]

Theorem 1 (The concentration)**.** *On the committed deck structure: (i) $A^\Gamma = A$ and $H^2(C_4; A) = 0$ (the norm degeneration $N = (1+\gamma)(1+\gamma^2) = 0$ with no invariants — the committed R1 law, re-derived at check 0008); (ii) $E^{0,2} = H^2(\Gamma; A)^{C_4}$ has rank exactly 12, and its θ-shaped part is exactly zero (checks 0012–0013): every invariant strip class is a non-θ cross-wedge tensor; (iii) consequently the map*

$$H^2(\mathcal{G}_\bullet; A) \longrightarrow H^1(C_4; H^1(\Gamma; A))$$

is injective*: every post-strip two-cell class is determined by its mixed-cell image.*

Proof. (i) is R1: $\ker(\gamma - 1) = 0$ on the content and $N = 0$, so $H^2(C_4; A) = \ker(\gamma - 1)/\operatorname{im} N = 0$. (ii) is the exact computation: the invariants of $\wedge^2\Gamma^* \otimes A$ under the cohomological action $\gamma \cdot (\phi \otimes a) = \gamma_A\, \phi(\gamma^{-1}\cdot, \gamma^{-1}\cdot)$ form a rank-12 lattice (the cross-wedge tensors (e_0, f_0)-free letters carry the invariants; the character content is $\wedge^2\Gamma^* = \mathbb{Q}(1) \oplus 6\, V_{-i} \oplus 15\, \mathbb{Q}(1)$ and $V_{-i} \otimes V_i \supset \mathbb{Q}(1)^{\oplus 2}$), while the θ-component $\theta \otimes a$ is invariant iff $(\gamma - 1)a = 0$, i.e. $a = 0$. (iii) exactness of the five-term sequence at $H^2(\mathcal{G}_\bullet; A)$: the kernel of the mixed-cell map is the image of $H^2(C_4; A^\Gamma) = 0$. □

Remark 2 (The honest finding)**.** *The naive expectation $E^{0,2} = 0$ is false: the mixed-glue of $\wedge^2\Gamma^* \otimes \mathbb{Q}$ under the deck produces 12 invariant tensors. The concentration does not run through strip-layer vanishing; it runs through the* injectivity *— and the injectivity is exactly what converts the mixed-layer torsion (§3) into the vanishing of the full class at the kill-scale.*

3 The mixed layer and the kill at scale $2m$ (T2) [PROVEN HERE + CERTIFICATE RECEIPT]

Theorem 3 (The structure of the mixed layer)**.** *$H^1(C_4; \mathrm{Hom}(\Gamma, A)) \cong (\mathbb{Z}/2)^6$: pure 2-torsion, order 64 (checks 0022–0023). More precisely the module $\mathrm{Hom}(\Gamma, A) \cong (\Gamma^* \otimes A)$ decomposes over $\mathbb{Q}$ as $(V_{-i} \otimes V_i) \oplus 6\, V_i$; the pair-0 part is $(\gamma_{\mathrm{act}} - 1)$-surjective (its anti-invariant lattice is the full image of the difference map — computed: the anti-invariants $\{F : \gamma F = -F\}$ are $\{d - a, -(b + c)\}$-spans, all reachable), and each of the six fixed-letter copies contributes exactly one $\mathbb{Z}/2$ (the committed R1 receipt $H^1(C_4; M) = \mathbb{Z}/2$, one copy per fixed letter).*

Corollary 4 (The kill)**.** *Every class μ in the mixed layer satisfies $2\mu = 0$ (the asserted Smith structure $(\mathbb{Z}/2)^6$, checks 0022–0023). The mixed-cell map of T1 is a homomorphism of abelian groups (the post-strip coefficients are abelian; the five-term's maps are group maps - standard), so* $\mathrm{image}(2[C_T^0]) = 2\,\mathrm{image}([C_T^0]) = 0$*, and by the injectivity (T1), $2\,[C_T^0] = 0$: the doubled transport's two-cell is a coboundary — cohomologically flat. Universally the mixed layer of a C_4-deck is 4-torsion (the classical bound; the honest boundary is the explicit glue module $\mathbb{Z}[x]/((x+1)(x^2+1))$ with $H^1 = \mathbb{Z}/4$, computed at checks 0025–0026 as the falsifiable control); on the committed model the Smith normal form excludes the glue: the kill is at $2m$, not $4m$.*

Composition with the committed stack. The descent criterion [DESCENT, committed 2026-10-23] converts the flat doubled transport into the descent of the $2m$-carrier; the two-center conduit (Lemma N + the universal two-center descent law, proven 2026-10-24) produces the cycle Z with $\mathrm{cl}(Z) = (-1)^{p-1}(p-1)!\cdot 2m\cdot\alpha$; the rational division by $2m(p-1)!$ exhibits α as a rational-algebraic class. The witness-level and input-sensitivity controls are §4 and §5; the remaining boundaries are §5. □

4 The strip-shape hypothesis resolved negatively (T3), and the witness (T4)

Theorem 5 (The strip-shape inconsistency; item (v))**.** *No nonzero ι-strip class is C_4-invariant: for $a \neq 0$ in the odd sector, $\gamma\cdot[\theta\otimes a] = [\theta\otimes\gamma_A a] \neq [\theta\otimes a]$ because $(\gamma_A - 1)a \neq 0$ and θ is symplectic-invariant (checks 0018–0019). Since the restriction of every $\mathcal{G}_\bullet$-class to the normal subgroup Γ lands in the invariant classes (the normality lemma, standard for normal subgroups; its normalization — the group law, the deck action, the differential — is verified end-to-end on the bar complex at the battery checks, and the concentration's own low end is verified directly by the derivation-level computation T1d: $H^1(\mathcal{G}_\bullet; A) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/2$, the five-term's exact prediction),* the strip-shape hypothesis of Round 24 is inconsistent in the odd sector*: the actual two-cell of the carrier cannot have the pure ι-strip shape. The correct shape is the* mixed-cell *shape: the identification link Λ_p's codomain $H^1(C_4; \iota(\pi_p(M)))$ is exactly the five-term's middle term; the link is the transgression, and its parity law (R5) is the 2-torsion refinement of §3.*

Theorem 6 (The witness)**.** *The $(A, -A)$ witness of AUDIT-22 F2, retyped at its content module (the ι-image of the Weil plane, R3), carries the mixed 1-cocycle $f_w(v) = \theta(v,x)\,W_R$ ($x = e_1$, a pinned deck-fixed letter - the six fixed letters carry six independent $\mathbb{Z}/2$'s): it satisfies $N f_w = 0$ exactly (the norm degeneration absorbs the content, check 0027), its class in $H^1(C_4; \mathrm{Hom}(\Gamma, A))$ is* nonzero at scale 1 *(check 0034: not $(\gamma_{\mathrm{act}} - 1)$-reachable in the* lattice *sense — the rational solutions carry the $1/2$-fractions that are exactly the 2-torsion obstruction, check 0034), and it* dies at scale 2 *(check 0033). Moreover the class is* realized*: the explicit two-cell cocycle $c_\nu((v,h),(w,k)) := \nu(h)(\gamma^h w)$ with ν the C_4-cocycle generated by f_w satisfies the full bar identity (checks 0038–0041: the strip zero, the deck cells zero, the mixed cell $\nu(1)$) - the witness's pair-level residue is carried by a genuine two-cell, and $d_2[f_w] = 0$ is discharged constructively (the 22-b F1 mode repaired: a lifting, not a retyping). The degenerate control (the 24-b M1 counterexample) is reproduced: contents in $(\gamma_A - 1)M$ carry no obstruction at any scale (check 0032). The parity structure of the reachability is the door-parity law itself: the even-coordinate-sum contents are reachable at scale 1, the odd rays W_R, W_I are not, and all die at 2 (checks 0030–0031) — R5 reproduced in the mixed layer, with the ray-uniformity receipt (ledger 097).*

5 The input-sensitivity (T5) and the boundaries on the face

The Chern-typing barrier (T5). The load-bearing (p, p)-step of the chain is the carrier's grade-p concentration: the p-th Chern character component of any coherent carrier is (p, p)-typed (Chern classes are algebraic; ch_p is a weight-p polynomial in them), so a pure-grade carrier $ch(E_+ \ominus E_-) \equiv (-1)^{p-1}(p-1)!\, m\, \alpha \pmod{\text{lower grades}}$ exists only for (p, p)-contents: the conductor output sector and the non-(p, p) letter sector are disjoint on the letter model (check 0043); the excluded-input run at the module level: the swap-typed content (the $(3, 1)$-analogue) is refused — its deck class survives, $H^2(C_4; \text{swap}) = \mathbb{Z}/2 \neq 0$ (check 0045), exactly where a non-Hodge content enters the chain. The killing chain provably refuses non-Hodge inputs at the carrier's construction — the Corollary-7 discipline, localized at the input step, with both receipts.

Item (iv) dissolved. This route uses no eigenbranch exchange: the deck is the rotation C_4, not the C_2-reflection; the rank-parity obstruction $r_0 = d + 1$ odd (the 21-c R1(3) receipt, first open case $d = 4$, $p = 2$, $r_0 = 5$) is vacuous on the rotation route. The kill is the H^1-torsion, not a branch split.

The boundaries that remain (on the face of this artifact).

(a) **The torsion boundary.** The committed D2/D3 strip reduces the two-cell to the traceless content; the nonabelian reconstruction of the flat doubled transport (the passage from the vanishing of the post-strip class to the multiplicative gauge correction of the transport) carries the central boundary $\partial \in H^3(\mathcal{G}_\bullet; \mu_2)$ ($\mu_2 = \{\pm I\}$ the center of $\mathrm{SL}(H_z)$). The strip is committed as proven at the scalar/trace level; the boundary class is the remaining piece of item (ii) at the nonabelian layer.

(b) **The fiber-centralizer condition.** The cochain model takes $A^\Gamma = A$ (the fiber periods act trivially on the content). The general input needs the irreducibility of the fiber transport (its centralizer in the traceless content is zero — the Stone–von Neumann-type rigidity of the θ-Heisenberg transport on the committed model). Where the centralizer is nonzero, the deck-layer term $H^2(C_4; A^\Gamma)$ need not vanish and the concentration must be re-run with the honest A^Γ.

(c) **The general-input deck (item (i) transformed).** Which (X, p, α) of the open sector carry the C_4-monodromy structure is the transport question — the ledger-085 boundary, unchanged. What this round adds: the boundary is now a *spectral-sequence* boundary: the input-axis transport is exactly the functoriality of the five-term sequence in the deck.

(d) **The glue at the general input.** The 2- vs 4-torsion distinction is decided by the Smith form of the mixed module; the committed model has no glue (2-torsion, kill $2m$); a general input with $(x+1)(x^2+1)$-glue kills at $4m$ — the rational quarter still closes, but the kill-scale is input-dependent.

6 The receipts

`research/experiments/two_cell_reduction_checks.py`: 125 checks, exit 0, deterministic double-run byte-identical, exact `Fraction`/integer arithmetic throughout (zero floats, zero complex), bounded systems (all Smith/elimination instances $\leq 56 \times 56$; the bar batteries bounded). After the audit repairs, the receipt structure: the asserted gates (T2's $(\mathbb{Z}/2)^6$ and the kill-scale, checks 0022–0023), the derivation-level five-term low end (T1d, checks 0016–0017: $H^1(\mathcal{G}_\bullet; A) = \mathbb{Z}^2 \oplus \mathbb{Z}/2$), the witness receipts (0027–0029: traceless, non-scalar, the pinned ray), the constructive lifting (T4-LIFT, checks

0038–0041), the excluded-input control (0045), and the hostile controls (the even-sector refusal, the negation module, the naive-cochain failure, the glue module, the trivial-content invariant detector). The committed anchors re-verified: $\gamma^4 = 1$, $\gamma \in \mathrm{Sp}(8, \mathbb{Z})$, $\gamma(W_R) = -W_I$, $H^1(C_4; M) = \mathbb{Z}/2$.

7 The scope line

At the committed deck structure: the two-cell reduction is *constructed* (the five-term concentration; AUDIT-22 F1's demand), the strip-shape hypothesis is *resolved negatively* (replaced by the mixed-cell shape; item (v)), the rank parity is *dissolved* (item (iv)), the witness is handled at its content class with the parity engine, and the input-sensitivity is localized at the Chern-typing barrier. The Clay statement is not claimed: the boundaries (a)–(d) are open mathematics. What is proven: at the committed model, the composition *carrier (Chern-typed input)* $\to$ *strip (D2/D3)* $\to$ *concentration (five-term)* $\to$ *kill (2m, SNF-decided)* $\to$ *descent (committed)* $\to$ *two-center conduit (proven)* $\to$ *rational algebraicity of* α is a closed chain of certificate-backed steps, and each remaining boundary is a named, falsifiable, input-axis object.